

Abdullah Rahat

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Summary

Software Engineer with experience building Agentic AI systems using FastAPI, LangChain, and LangGraph, including RAG pipelines, asynchronous workflows, and observability tooling. Background in backend development for ERP systems and cross-platform Flutter applications. Published researcher in deep learning model compression.

Technical Skills

Languages: Python, SQL, Java, Dart

Frameworks: FastAPI, Spring Boot, Flutter (Riverpod, GoRouter, Hive, Dio)

Databases: PostgreSQL, Pgvector, Redis

AI/ML: LangChain, LangGraph, RAG, Langfuse, OCR, XGBoost

Tools: Docker, Git, Celery, Postman, Prometheus, Loki, Grafana, Linux

Experience

Software Engineer I

Oct 2025 – Present

Wyze Tech Ltd

Dhaka, Bangladesh

- Built a Flutter application consuming FastAPI backend services, following clean architecture and modular design with Riverpod, GoRouter, Hive, and Dio.
- Developed Agentic AI systems using FastAPI, LangChain, and LangGraph with multi-step tool execution, integrating Langfuse for observability.
- Implemented asynchronous workflows using Redis and Celery, and configured Prometheus, Loki, and Grafana for logging and tracing.
- Implemented RAG pipelines using Pgvector and external knowledge sources to generate contextual responses.

Software Engineer

Feb 2025 – Sep 2025

Giga Tech Ltd

Dhaka, Bangladesh

- Developed and maintained backend services for ERP systems using FastAPI and PostgreSQL.
- Designed REST APIs for inventory, order, and user workflows with optimized data handling.
- Built and deployed a containerized OCR service for ERP document processing.
- Implemented report enhancements, production bug fixes, and transaction code maintenance for an ERP system supporting one million users.
- Supported functional consultants by analyzing business requirements and database tables to assist in issue investigation and validation.

Projects

Blog Platform API (Spring Boot)

GitHub

- Engineered a production-ready blog platform API with JWT authentication, role-based authorization, and OTP-based email verification.
- Architected a scalable backend with dependency injection, validation, and service-layer separation using PostgreSQL and Hibernate.
- Integrated Gemini API for AI-powered content generation and containerized services with Docker; documented via OpenAPI/Swagger.

Email Generation Assistant (FastAPI, LangGraph)

GitHub

- Built an email generation system using LangGraph and OpenRouter with dual LLM models for response comparison.
- Implemented structured outputs with Pydantic, a modular service-layer architecture, and SQLAlchemy-based persistence.
- Developed an evaluation pipeline measuring fact coverage, tone alignment, and structure quality, producing automated CSV/JSON reports.

MediAI (FastAPI, LangGraph)

GitHub

- Built an AI-powered healthcare assistant integrating clinical risk prediction, patient triage, and structured medical note generation.
- Developed an XGBoost-based risk stratification model and LLM-driven specialty routing with structured JSON outputs.
- Designed a modular architecture with environment-based configuration and an interactive API documentation interface.

ERPNext Agentic AI System

GitHub

- Built an LLM-powered agent integrating ERPNext via Frappe APIs for natural language-driven operations.
- Implemented tool-based execution with schema enforcement, failure handling, and persistent memory with summarization.
- Supported hybrid LLMs (Ollama, Gemini) and integrated Langfuse for observability.

ReAct Graph AI Agent

GitHub

- Engineered a graph-based LLM agent using LangGraph with conditional routing and stateful execution.
- Implemented ReAct reasoning with dynamic tool invocation across search, ArXiv, and Wikipedia sources.
- Integrated multi-model backends (Gemini, Ollama) with a FastAPI layer for scalable, asynchronous interaction.

Emergency Connect (Flutter)

GitHub

- Developed a cross-platform emergency assistance application with local JSON-based data management.
- Implemented platform-specific support for Android, iOS, Linux, macOS, and Windows using reusable widgets.

To-Do API (Spring Boot)

GitHub

- Developed a task management REST API with layered architecture, DTO mapping, and Docker-based deployment.

Task Manager CLI (Java)

GitHub

- Developed a CLI-based task manager applying OOP principles and clean modular design for task operations.

Education

Bachelor of Science in Computer Science

2020 – 2024

BRAC University, CGPA: 3.45

Thesis: *Knowledge Distillation and Weight Pruning for Two-step Compression of ConvNets in Rice Leaf Disease Classification*. Published in NSysS Conference, November 2024. DOI Link

Relevant Coursework: Pattern Recognition, Neural Networks, Machine Learning, Image Processing, Data Science

Certifications

Python (Basic) – HackerRank, Dec 2024	Certificate
Java (Basic) – HackerRank, Dec 2024	Certificate
SQL (Basic) – HackerRank, Dec 2024	Certificate

Extra-Curricular Activities

- Achieved recognition in the National Math Olympiad, National Science Olympiad, and other competitive events. Certificate
- Ranked in the BUCC Intra Department Coding Contest.
- Served as an Executive Member of the BRAC University Computer Club.